

TABLE 1. Compositions of standard media^a

Me- dium	Component															
	Mineral 1 ^b (ml)	Mineral 2 ^c (ml)	Mineral 3 ^d (ml)	NaCl (g)	NH ₄ Cl (g)	Trace min- erals ^e (ml)	Trace vita- mins ^f (ml)	FeSO ₄ · 7H ₂ O (g)	Fe(NH ₄) ₂ (SO ₄) ₂ · 7H ₂ O (g)	NaHCO ₃ (g)	Sodium acetate (g)	Sodium formate (g)	Yeast extract (Difco) (g)	Trypti- case (BBL)	L-Cys- teine hydro- chloride· H ₂ O (g)	Na ₂ S· 9H ₂ O (g)
1	50	50				10	10	0.002		5.0	2.5	2.5	2.0	2.0	0.5	0.5
2		25			1.25	10	10		0.02	7.5					0.6	0.6
3			500	18		10	10		0.002	5.0	1.0		2.0	2.0	0.5	0.5

^a Ingredients are added to distilled water to give a final volume of 1 liter. Cysteine and Na₂S are added after boiling the medium under an 80% N₂–20% CO₂ gas mixture, the final gas phase of tubed medium being an 80% H₂–20% CO₂ gas mixture at two atmospheres of pressure.

^b Contains 6 g of K₂HPO₄ per liter of distilled water.

^c Contains, in grams per liter of distilled water: KH₂PO₄, 6; (NH₄)₂SO₄, 6; NaCl, 12; MgSO₄·7H₂O, 2.6; CaCl₂·2H₂O, 0.16.

^d Contains, in grams per liter of distilled water: KCl, 0.67; MgCl₂·2H₂O, 5.5; MgSO₄·7H₂O, 6.9; NH₄Cl, 0.5; CaCl₂·2H₂O, 0.28; K₂HPO₄, 0.28.

^e Contains, in grams per liter of distilled water (pH to 7.0 with KOH): nitrilotriacetic acid, 1.5; MgSO₄·7H₂O, 3.0; MnSO₄·2H₂O, 0.5; NaCl, 1.0; FeSO₄·7H₂O, 0.1; CoSO₄ or CoCl₂, 0.1; CaCl₂·2H₂O, 0.1; ZnSO₄, 0.1; CuSO₄·5H₂O, 0.01; AlK(SO₄)₂, 0.01; H₃BO₃, 0.01; Na₂MoO₄·2H₂O, 0.01. Dissolve nitrilotriacetic acid with KOH to pH 6.5; then proceed to add minerals.

^f Contains, in milligrams per liter of distilled water: biotin, 2; folic acid, 2; pyridoxine hydrochloride, 10; thiamine hydrochloride, 5; riboflavin, 5; nicotinic acid, 5; DL-calcium pantothenate, 5; vitamin B₁₂, 0.1; *p*-aminobenzoic acid, 5; lipoic acid, 5.

For *M. ruminantium* M1 (type strain) Medium 1 has some modifications:

Only 2.5 g NaHCO₃

0.5 Valeric acid

0.5 Isovaleric acid

0.5 *a*-methylbutyrate

0.5 isobutyrate

0.0001 2-mercaptoethanesulfonic acid

Media composition for *Methanobrevibacter ruminantium* M1

Substance	g L ⁻¹ / mL	mM	comment
Mineral 1	50		
Mineral 2	50		
Trace minerals	10		
Trace vitamins	1		
FeSO ₄ x 7 H ₂ O	0.002		
NaHCO ₃	2.5		
Na-acetate	2.5		
Na-formate	2.5		
Yeast extract	2		
Peptone from Casein, tryptic digested	2		
L-Cys-HCl x 9 H ₂ O	0.5		Added after autoclavation Stock solution 50 g L ⁻¹
Na ₂ S x 9 H ₂ O	0.5		Added after autoclavation Stock solution 100 g L ⁻¹

Dissolve in H₂O besides NaHCO₃ and Vitamins and bring to a boil. Add NaHCO₃ and gas with N₂ + CO₂

Mineral 1

Substance	g L ⁻¹ / mL	mM	comment
K ₂ HPO ₄	6 g		

Mineral 2

Substance	g L ⁻¹ / mL	mM	comment
KH ₂ PO ₄	6		
(NH ₄) ₂ SO ₄	6		
NaCl	12		
MgSO ₄ x 7 H ₂ O	2.6		
CaCl ₂ x 2 H ₂ O	0.16		

Trace minerals

Substance	g L ⁻¹ / mL	mM	comment
Nitrolotriacetic acid	1.5		
MgSO ₄ x 7 H ₂ O	3		
MnSO ₄ x 2 H ₂ O	0.5		
NaCl	1		
FeSO ₄ x 7 H ₂ O	0.1		
CoSO ₄ / CoCl ₂	0.1		
CaCl ₂ x 2 H ₂ O	0.1		
ZnSO ₄	0.1		
CuSO ₄ x 5 H ₂ O	0.01		
AlK(SO ₄) ₂	0.01		
H ₃ BO ₃	0.01		
Na ₂ MoO ₄ x 2 H ₂ O	0.01		

Dissolve nitrolotriacetic acid in KOH to pH 6.5, then add minerals. pH to 7.0 with KOH

Trace vitamins 1000x

Substance	mg L ⁻¹ / mL	mM	comment
Biotin	20		
Folic acid	20		
Pyridoxin-HCl	100		
Thiamine-HCl	50		
Riboflavin	50		
Nicotinic acid	50		
DL-Ca-panthothenate	50		
Vit B ₁₂	1		
p-aminobenzoic acid	50		
Lipoic acid	50		

Fatty acid mixture

Substance	g L ⁻¹ / mL	mM	comment
Valeric acid	0.5		
Isovaleric acid	0.5		
a-methylbutyric acid	0.5		
Isobutyric acid	0.5		

Coenzyme M

Substance	g L ⁻¹ / mL	mM	comment
2-mercaptoethanesulfonic acid	0.0001		